

## **Exercises on Production Possibilities Model**

### Question 1

Suppose there is a country that can produce two goods: X and Y. The country has 100 machines and 50 workers. Each machine can produce 2 units of X or 4 units of Y. Each worker can produce 6 units of X or 6 units of Y.

- a. Who has a comparative advantage in producing X, workers or machines?
- b. Who has a comparative advantage in producing Y, workers or machines?
- c. Please draw a production possibilities frontier of this country.
- d. Suppose that this country needs to consume 400 units of X. In that case how many units of Y will be produced?

### Question 2

Suppose there are two independent countries Zulu and Wulu. Those countries consume two goods, guns (X) and butter (Y).

Assume that in Zulu there are 100 workers and each of them can produce 1 unit of guns or 1 unit of butter. Assume that in Wulu there are 400 workers and each of them can produce 2 units of guns or 1 unit of butter.

- a. Please draw a production possibilities frontier of each of these countries.
- b. It is known that each country needs to consume 25 guns for self-defence. How many units of butter will be consumed by each country?
- c. Now assume that there is a world trade, such that 1 unit of butter is worth 1.5 units of guns. Please draw a production possibilities frontier of each of these countries.
- d. If each country still needs 25 units of guns, what will be the consumption of butter of each of these countries?
- e. Please explain whether trade is profitable for each of these countries.

### Question 3

Explain the following sentence: After many years of a heavy military production on expense of technological development of other fields, many countries are not expected to have a sustainable economic growth.